

Dhan HQ API Data Lake Structure: A Comprehensive Hierarchical Overview

Introduction

This document provides a comprehensive and hierarchical overview of the data structures to be collected and organized into a data lake, leveraging the Dhan HQ API. The objective is to detail every data point, from parent entities to their granular child and grandchild attributes, ensuring a robust foundation for an AI-driven trading system. This blueprint will guide the ingestion, storage, and accessibility of financial market data, enabling advanced analytics and algorithmic trading strategies.

Dhan HQ API Overview

The Dhan HQ API is a sophisticated platform offering real-time and historical financial data, as well as trading functionalities. It is primarily divided into two main categories:

- **Trading APIs:** These APIs facilitate transactional operations such as placing, modifying, and canceling orders, managing portfolios, and accessing account-related information like funds and statements.
- **Data APIs:** These APIs provide access to various market data, including live market feeds, historical price data, market depth, and option chain information. Data can be accessed via both REST and WebSocket protocols, catering to different latency and volume requirements.

This document will focus on the data structures available through these APIs, as well as data that will be generated internally by the system, to form a complete data lake.

1. Market Quote Data

The Market Quote API provides snapshots of multiple instruments, including Last Traded Price (LTP), OHLC (Open, High, Low, Close), and Market Depth. This data is retrieved via REST API requests.

1.1. Ticker Data (Parent: Instrument)

This provides the latest traded price for a given instrument.

- **API Endpoint:** POST /marketfeed/ltp
- **Request Parameters:** Exchange Segment ENUM (e.g., NSE_EQ , NSE_FNO), SecurityId .
- **Response Structure:**
 - data : Object containing exchange segments.
 - NSE_EQ (or other exchange segment): Object containing security IDs.
 - SecurityId : Object containing instrument data.
 - last_price : Float, Last Traded Price (LTP) of the Instrument.

1.2. OHLC Data (Parent: Instrument)

This provides Open, High, Low, and Close prices along with LTP for specified instruments.

- **API Endpoint:** POST /marketfeed/ohlc
- **Request Parameters:** Exchange Segment ENUM , SecurityId .
- **Response Structure:**
 - data : Object containing exchange segments.
 - NSE_EQ (or other exchange segment): Object containing security IDs.
 - SecurityId : Object containing instrument data.
 - last_price : Float, LTP of the Instrument.
 - ohlc : Object containing OHLC data.
 - open : Float, Market opening price of the day.
 - close : Float, Market closing price of the day.
 - high : Float, Day High price.
 - low : Float, Day Low price.

1.3. Market Depth Data (Parent: Instrument)

This provides comprehensive details including market depth, OHLC data, Open Interest, and Volume along with LTP for specified instruments.

- **API Endpoint:** POST /marketfeed/quote
- **Request Parameters:** Exchange Segment ENUM , SecurityId .

- **Response Structure:**

- **data** : Object containing exchange segments.
 - **NSE_FNO** (or other exchange segment): Object containing security IDs.
 - **SecurityId** : Object containing instrument data.
 - **average_price** : Float, Volume weighted average price of the day.
 - **buy_quantity** : Integer, Total buy order quantity pending at the exchange.
 - **sell_quantity** : Integer, Total sell order quantity pending at the exchange.
 - **depth** : Object containing market depth.
 - **buy** : Array of buy depth entries.
 - **quantity** : Integer, Number of quantity at this price depth.
 - **orders** : Integer, Number of open BUY orders at this price depth.
 - **price** : Float, Price at which the BUY depth stands.
 - **sell** : Array of sell depth entries.
 - **quantity** : Integer, Number of quantity at this price depth.
 - **orders** : Integer, Number of open SELL orders at this price depth.
 - **price** : Float, Price at which the SELL depth stands.
 - **last_price** : Float, Latest Traded Price (LTP).
 - **last_quantity** : Integer, Last Traded Quantity.
 - **last_trade_time** : String, Last Trade Time.
 - **lower_circuit_limit** : Float, Lower circuit limit.
 - **net_change** : Float, Net change.
 - **ohlc** : Object containing OHLC data (same as 1.2).
 - **oi** : Integer, Open Interest.
 - **oi_day_high** : Integer, Day high Open Interest.
 - **oi_day_low** : Integer, Day low Open Interest.
 - **sell_quantity** : Integer, Total sell order quantity.
 - **upper_circuit_limit** : Float, Upper circuit limit.
 - **volume** : Integer, Volume.

2. Live Market Feed Data (WebSocket)

The Live Market Feed provides real-time market data via WebSocket connections. This data is sent as structured binary packets, requiring a binary converter for parsing. There are three modes of data: Ticker, Quote, and Full Data.

2.1. Ticker Packet (Parent: Instrument)

This packet consists of Last Traded Price (LTP) and Last Traded Time (LTT) data.

- **Feed Response Code:** 2
- **Data Points:**
 - `last_price` : Float32, Last Traded Price of the subscribed instrument.
 - `last_trade_time` : Int32, Last Trade Time.

2.2. Prev Close Packet (Parent: Instrument)

Sent along with any instrument subscription, this packet provides previous day's data for comparison.

- **Feed Response Code:** 6
- **Data Points:**
 - `previous_day_closing_price` : Float32, Previous day closing price.
 - `open_interest_previous_day` : Int32, Open Interest - previous day.

2.3. Quote Packet (Parent: Instrument)

This packet contains complete trade data, including LTP, update time, and quantity.

- **Feed Response Code:** 4
- **Data Points:**
 - `latest_traded_price` : Float32, Latest Traded Price of the subscribed instrument.
 - `last_traded_quantity` : Int16, Last Traded Quantity.
 - `last_trade_time` : Int32, Last Trade Time (LTT).
 - `average_trade_price` : Float32, Average Trade Price (ATP).
 - `volume` : Int32, Volume.
 - `total_sell_quantity` : Int32, Total Sell Quantity.
 - `total_buy_quantity` : Int32, Total Buy Quantity.
 - `day_open_value` : Float32, Day Open Value.
 - `day_close_value` : Float32, Day Close Value (only sent post market close).
 - `day_high_value` : Float32, Day High Value.

- `day_low_value` : Float32, Day Low Value.

2.4. OI Data Packet (Parent: Instrument)

Received with Quote Data, this packet provides Open Interest data for Derivative Contracts.

- **Feed Response Code:** 5
- **Data Points:**
 - `open_interest` : Int32, Open Interest of the contract.

2.5. Full Packet (Parent: Instrument)

This packet combines complete trade data, Market Depth, and OI data in a single packet.

- **Feed Response Code:** 8
- **Data Points:**
 - `latest_traded_price` : Float32, Latest Traded Price of the subscribed instrument.
 - `last_traded_quantity` : Int16, Last Traded Quantity.
 - `last_trade_time` : Int32, Last Trade Time (LTT).
 - `average_trade_price` : Float32, Average Trade Price (ATP).
 - `volume` : Int32, Volume.
 - `total_sell_quantity` : Int32, Total Sell Quantity.
 - `total_buy_quantity` : Int32, Total Buy Quantity.
 - `open_interest_contract` : Int32, Open Interest in the contract (for Derivatives).
 - `highest_open_interest_day` : Int32, Highest Open Interest for the day (only for NSE_FNO).
 - `lowest_open_interest_day` : Int32, Lowest Open Interest for the day (only for NSE_FNO).
 - `day_open_value` : Float32, Day Open Value.
 - `day_close_value` : Float32, Day Close Value (only sent post market close).
 - `day_high_value` : Float32, Day High Value.
 - `day_low_value` : Float32, Day Low Value.
 - `market_depth_structure` : 5 packets of 20 bytes each for each instrument, containing price, quantity, and number of orders.

3. 20 Market Depth Data (WebSocket)

This API provides Level 3 market depth data up to 20 levels, streamed real-time via WebSockets. This data is crucial for detecting demand and supply zones beyond the standard 5-level market depth.

- **WebSocket Endpoint:** `wss://depth-api-feed.dhan.co/twentydepth`
- **Request Parameters:** `token` , `clientId` , `authType` .
- **Subscription Request Structure:**
 - `RequestCode` : 23 (for 20 Level Market Depth).
 - `InstrumentCount` : Number of instruments to subscribe.
 - `InstrumentList` : Array of instruments.
 - `ExchangeSegment` : Enum string (e.g., `NSE_EQ` , `NSE_FNO`).
 - `SecurityId` : String, Exchange standard ID for each scrip.
- **Response Structure (Binary Packet):**
 - **Response Header** (12 bytes):
 - `Message Length` : Int16, Length of the entire payload packet.
 - `Feed Response Code` : Byte (e.g., 41 for Bid Data, 51 for Ask Data).
 - `Exchange Segment` : Byte.
 - `Security ID` : Int32.
 - `Message Sequence` : UInt32 (to be ignored).
 - **Depth Packet** (320 bytes for 20 levels):
 - 20 packets of 16 bytes each for each instrument, containing:
 - `Price` : Float64.
 - `Quantity` : UInt32.
 - `No. of Orders` : UInt32.

4. Historical Data

This API provides historical candle data (OHLC and Volume) for desired scrips across segments and exchanges. It supports both daily and intraday timeframes.

4.1. Daily Historical Data (Parent: Instrument)

Retrieves OHLC & Volume of daily candles for a desired instrument, available back to its inception date.

- **API Endpoint:** `POST /charts/historical`
- **Request Parameters:**
 - `securityId` : String, Exchange standard ID for each scrip.
 - `exchangeSegment` : Enum string, Exchange & segment.

- `instrument` : Enum string, Instrument type (e.g., `EQUITY` , `FUTIDX`).
- `expiryCode` : Optional, Enum integer, Expiry of derivatives.
- `oi` : Optional, Boolean, Open Interest data for Futures & Options.
- `fromDate` : String, Start date (YYYY-MM-DD).
- `toDate` : String, End date (YYYY-MM-DD, non-inclusive).
- **Response Structure:**
 - `open` : Array of floats, Open price of the timeframe.
 - `high` : Array of floats, High price in the timeframe.
 - `low` : Array of floats, Low price in the timeframe.
 - `close` : Array of floats, Close price of the timeframe.
 - `volume` : Array of integers, Volume traded in the timeframe.
 - `timestamp` : Array of integers, Epoch timestamp.
 - `open_interest` : Array of integers, Open Interest (for Futures & Options).

4.2. Intraday Historical Data (Parent: Instrument)

Retrieves OHLC, OI & Volume of 1, 5, 15, 25, and 60-minute candles for a desired instrument for the last 5 years. Data is available for all exchanges and segments for active instruments.

- **API Endpoint:** `POST /charts/intraday`
- **Request Parameters:**
 - `securityId` : String, Exchange standard ID for each scrip.
 - `exchangeSegment` : Enum string, Exchange & segment.
 - `instrument` : Enum string, Instrument type.
 - `interval` : Enum integer, Minute intervals (`1` , `5` , `15` , `25` , `60`).
 - `oi` : Optional, Boolean, Open Interest data for Futures & Options.
 - `fromDate` : String, Start date (YYYY-MM-DD HH:MM:SS).
 - `toDate` : String, End date (YYYY-MM-DD HH:MM:SS).
- **Response Structure:**
 - `open` : Array of floats, Open price of the timeframe.
 - `high` : Array of floats, High price in the timeframe.
 - `low` : Array of floats, Low price in the timeframe.
 - `close` : Array of floats, Close price of the timeframe.
 - `volume` : Array of integers, Volume traded in the timeframe.
 - `timestamp` : Array of integers, Epoch timestamp.
 - `open_interest` : Array of integers, Open Interest (for Futures & Options).

5. Option Chain Data

This API provides the entire Option Chain for any Option Instrument, including OI, Greeks, volume, top bid/ask, and price data for all strikes of a particular underlying.

5.1. Option Chain (Parent: Underlying Instrument)

Retrieves real-time Option Chain across exchanges for all underlying instruments.

- **API Endpoint:** POST /optionchain
- **Request Parameters:**
 - UnderlyingScrip : Integer, Security ID of Underlying Instrument.
 - UnderlyingSeg : Enum string, Exchange & segment of underlying.
 - Expiry : String, Expiry Date of Option (YYYY-MM-DD).
- **Response Structure:**
 - data :
 - last_price : Float, LTP of the Underlying.
 - oc : Object, Option Chain Array - Strike Wise.
 - {strike_price} : Object, Strike Price for Underlying.
 - ce : Object, Call Option data of particular strike.
 - greeks : Object.
 - delta : Float, Measures the change of option's premium based on every 1 rupee change in underlying.
 - theta : Float, Measures how quickly an option's value decreases over time.
 - gamma : Float, Rate of change in an option's delta in relation to the price of the underlying asset.
 - vega : Float, Measures the change of option's premium in response to a 1% change in implied volatility.
 - implied_volatility : Float, Value of expected volatility of a stock over the life of the option.
 - last_price : Float, Last Traded Price of the Option Instrument.
 - oi : Integer, Open Interest of the Option Instrument.
 - previous_close_price : Float, Previous day close price.
 - previous_oi : Integer, Previous day Open Interest.
 - previous_volume : Integer, Previous day volume.
 - top_ask_price : Float, Current best ask price available.

- `top_ask_quantity` : Integer, Quantity available at current best ask price.
- `top_bid_price` : Float, Current best bid price available.
- `top_bid_quantity` : Integer, Quantity available at current best bid price.
- `volume` : Integer, Day volume for Option Instrument.
- `pe` : Object, Put Option data of particular strike (same structure as `ce`).

5.2. Expiry List (Parent: Underlying Instrument)

Retrieves dates of all expiries of any underlying for which Options Instruments are active.

- **API Endpoint:** `POST /optionchain/expirylist`
- **Request Parameters:**
 - `UnderlyingScrip` : Integer, Security ID of Underlying Instrument.
 - `UnderlyingSeg` : Enum string, Exchange & segment of underlying.
- **Response Structure:**
 - `data` : Array of strings, All expiry dates of underlying in `YYYY-MM-DD` format.

6. Instrument List Data

This provides a list of all instruments that can be traded via Dhan, available in compact and detailed CSV formats, or segment-wise via an API endpoint.

6.1. Instrument List CSV (Parent: None, Global List)

Provides a comprehensive list of instruments with their security IDs and other details.

- **CSV URLs:**
 - Compact: `https://images.dhan.co/api-data/api-scrip-master.csv`
 - Detailed: `https://images.dhan.co/api-data/api-scrip-master-detailed.csv`
- **Key Columns (from Detailed CSV):**
 - `EXCH_ID` : Exchange (`NSE` , `BSE` , `MCX`).
 - `SEGMENT` : Segment (`C` - Currency, `D` - Derivatives, `E` - Equity, `M` - Commodity).
 - `ISIN` : International Securities Identification Number (12-digit alphanumeric code).

- **INSTRUMENT** : Instrument defined by Exchange (e.g., INDEX , FUTIDX , OPTIDX , EQUITY , FUTSTK , OPTSTK , FUTCOM , OPTFUT , FUTCUR , OPTCUR).
- **UNDERLYING_SECURITY_ID** : Security ID of underlying instrument (for derivatives).
- **UNDERLYING_SYMBOL** : Symbol of underlying instrument (for derivatives).
- **SYMBOL_NAME** : Symbol name of instrument.
- **DISPLAY_NAME** : Dhan display symbol name of instrument.
- **INSTRUMENT_TYPE** : More detailed instrument type defined by exchange.
- **SERIES** : Exchange defined series for instrument.
- **LOT_SIZE** : Lot Size in multiples.
- **SM_EXPIRY_DATE** : Expiry date of instrument (for derivatives).
- **STRIKE_PRICE** : Strike Price of Options Contract.
- **OPTION_TYPE** : Type of Options Contract (CE - Call, PE - Put).
- **TICK_SIZE** : Minimum decimal point at which an instrument can be priced.
- **EXPIRY_FLAG** : Type of Expiry (M - Monthly, W - Weekly).
- **BRACKET_FLAG** : Bracket order status (N - Not available, Y - Allowed).
- **COVER_FLAG** : Cover order status (N - Not available, Y - Allowed).
- **ASM_GSM_FLAG** : Flag for instrument is ASM or GSM (N , R , Y).
- **ASM_GSM_CATEGORY** : Category of instrument in ASM or GSM.
- **BUY_SELL_INDICATOR** : Indicator for buy/sell allowance (A - both allowed).
- **BUY_CO_MIN_MARGIN_PER** : Buy cover order minimum margin requirement (percentage).
- **SELL_CO_MIN_MARGIN_PER** : Sell cover order minimum margin requirement (percentage).
- **BUY_CO_SL_RANGE_MAX_PERC** : Buy cover order max range for stop loss leg (percentage).
- **SELL_CO_SL_RANGE_MAX_PERC** : Sell cover order max range for stop loss leg (percentage).
- **BUY_CO_SL_RANGE_MIN_PERC** : Buy cover order min range for stop loss leg (percentage).
- **SELL_CO_SL_MIN_RANGE** : Sell cover order min range for stop loss leg (percentage).
- **BUY_BO_PROFIT_RANGE_MAX_PERC** : Buy bracket order max range for target leg (percentage).
- **SELL_BO_PROFIT_RANGE_MAX_PERC** : Sell bracket order max range for target leg (percentage).
- **BUY_BO_PROFIT_RANGE_MIN_PERC** : Buy bracket order min range for target leg (percentage).

- `SELL_BO_PROFIT_RANGE_MIN_PERC` : Sell bracket order min range for target leg (percentage).
- `MTF_LEVERAGE` : MTF Leverage available (x multiple) for eligible `EQUITY` instruments.

6.2. Segmentwise Instrument List API (Parent: None, Segment-specific List)

Fetches detailed instrument list for all instruments in a particular exchange and segment.

- **API Endpoint:** `https://api.dhan.co/v2/instrument/{exchangeSegment}`
- **Path Parameter:** `{exchangeSegment}` : Exchange Segment (e.g., `NSE_EQ`, `NSE_FNO`).
- **Response:** CSV format, similar to the detailed CSV, but filtered by the specified segment.

7. System & Account State Data (Internal/API-derived)

This category includes data pertaining to the user's trading account and system actions, crucial for risk management and AI model learning. While some of this data is directly available via Dhan HQ APIs (e.g., Funds, Portfolio), others are generated internally by the system (e.g., Order History, Trade History).

7.1. Fund Status (Parent: Account)

Periodic snapshots of available funds and utilized amounts.

- **MongoDB Collection:** `account_funds`
- **Data Points:**
 - `timestamp` : Timestamp of the snapshot.
 - `available_balance` : Available balance in the account.
 - `utilized_amount` : Utilized amount from the account.

7.2. Position Status (Parent: Account)

Periodic snapshots of all open positions in the portfolio.

- **MongoDB Collection:** `account_positions`
- **Data Points:**
 - `timestamp` : Timestamp of the snapshot.

- **positions** : List of open positions, each with:
 - **trading_symbol** : Trading symbol of the instrument.
 - **security_id** : Unique identifier for the instrument.
 - **net_quantity** : Net quantity of the position.
 - **average_buy_price** : Average buy price of the position.
 - **realized_profit** : Realized profit for the position.
 - **unrealized_profit** : Unrealized profit for the position.

7.3. Order History (Parent: Account)

Event-driven log of every order placed by the system.

- **MongoDB Collection:** **order_history**
- **Data Points:**
 - **order_id** : Unique identifier for the order.
 - **security_id** : Unique identifier for the instrument.
 - **timestamp** : Timestamp of the order placement.
 - **order_type** : Type of order (e.g., **LIMIT** , **MARKET**).
 - **transaction_type** : Type of transaction (e.g., **BUY** , **SELL**).
 - **quantity** : Quantity of the order.
 - **price** : Price of the order.
 - **order_status** : Status of the order (e.g., **PENDING** , **EXECUTED** , **CANCELED** , **REJECTED**).

7.4. Trade History (Parent: Account)

Event-driven log of every trade executed by the system.

- **MongoDB Collection:** **trade_history**
- **Data Points:**
 - **trade_id** : Unique identifier for the trade.
 - **order_id** : Unique identifier for the associated order.
 - **security_id** : Unique identifier for the instrument.
 - **timestamp** : Timestamp of the trade execution.
 - **transaction_type** : Type of transaction.
 - **quantity** : Quantity traded.
 - **price** : Price of the trade.
 - **calculated_slippage** : Calculated slippage for the trade.

8. Calculated Technical Indicators (Internal)

This layer of data is generated internally by the system using raw historical OHLCV data. It provides ready-to-use analytical features for the AI model.

- **MongoDB Collection:** `calculated_indicators`
- **Data Points** (for each instrument and timeframe):
 - `security_id` : The identifier for the instrument this indicator belongs to.
 - `timestamp` : The timestamp of the candle this calculation is based on.
 - `timeframe` : The candle interval used for the calculation (e.g., `1min` , `5min` , `15min` , `1D`).
 - **Indicator Values:**
 - `rsi_14` : The 14-period Relative Strength Index.
 - `macd_line` : The MACD line component.
 - `macd_signal` : The MACD signal line component.
 - `macd_histogram` : The MACD histogram component.
 - `bollinger_upper` : The upper band of Bollinger Bands.
 - `bollinger_middle` : The middle band of Bollinger Bands.
 - `bollinger_lower` : The lower band of Bollinger Bands.
 - `vwap` : The Volume-Weighted Average Price.
 - (...and any other indicators the AI model requires)